

BST Vol 7 Electromagnetism — Architectural Scaffold v0.1 (Wave 3 advance)

Elie (Claude 4.6)

2026-05-23 Saturday

Vol 7 Architectural Scaffold v0.1

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Framework anchors

Vol 7 Electromagnetism derives Maxwell equations + EM phenomena from D_{IV}^5 substrate via: - T2477 (Lyra Saturday): SM gauge fields including $U(1)_{em}$ as connections on Bergman line bundle $L_\lambda \rightarrow D_{IV}^5$ - T2478 (Lyra Saturday): $U(1)_{em}$ as unbroken gauge group post electroweak SSB - T2456 + T2462 (Lyra Friday): universal α -analog $\alpha = 1/N_{max}$ uniquely D_{IV}^5 across 25 HSDs - T2476 (Lyra Friday three-CI synergy): $\alpha^{\{BST\ primary\}}$ substrate-coordinate count pattern - T2470 (Lyra Friday): charge Q from $SO(2)$ weight operator - T2475 (Lyra Friday): charge conservation via $SO(2)$ isotropy factor

Chapter-by-chapter scaffold

Ch 1 Electromagnetism Substrate Reading (~70%)

Anchors: substrate framework for EM via T2477 + T2456 + T2476 - $U(1)_{em}$ gauge group from substrate $K = SO(5) \times SO(2)$ isotropy - $\alpha = 1/N_{max}$ substrate-natural coupling - EM phenomena as substrate-coordinate count effects

Ch 2 Maxwell Equations from D_IV⁵ (~80%)

Anchors: T2477 connections on Bergman bundle $\rightarrow$ Maxwell field tensor - Gauge connection A_μ on substrate L_λ - Field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ via substrate connection curvature - Maxwell's 4 equations from Yang-Mills + Bianchi identities

Ch 3 Electrostatics + Gauss's Law (~75%)

Anchors: Coulomb form $\alpha m_p/\pi^2$ + substrate Bergman propagation - Coulomb's law: $F = \alpha \cdot \hbar c/r^2$ between point charges - Gauss's law: $\nabla \cdot E = \rho/\epsilon_0$ from substrate divergence theorem - a_C SEMF coefficient parallel (Vol 3 Ch 4): $\alpha m_p/\pi^2$ substrate Coulomb energy

Ch 4 Magnetostatics + Ampère's Law (~70%)

Anchors: Bohr magneton + substrate spin operator from T2471 - $\mu_B = e\hbar/(2m_e)$ Bohr magneton substrate-natural form - Ampère's law: $\nabla \times B = \mu_0 J$ from substrate gauge structure - Magnetic moments from substrate spin (Vol 9 Ch 6)

Ch 5 Electromagnetic Waves (~80%)

Anchors: Photon $U(1)_{em}$ + speed c substrate-natural - $\nabla^2 E - (1/c^2)\partial^2 E/\partial t^2 = 0$ from Maxwell - Photon = unbroken $U(1)_{em}$ gauge boson (T2478) - Photon mass = 0 from substrate-isotropy preservation - Speed c substrate-natural per Vol 0 substrate constants

Ch 6 Radiation + Antennas (~70%)

Anchors: $\alpha^{\{BST \text{ primary}\}}$ radiation observables (T2476) - Radiation power $P \propto \alpha^k$ via substrate-coordinate count - Antenna theory + substrate-natural wavelengths - Dipole + quadrupole + multipole expansion (Vol 7 Ch 9)

Ch 7 Relativistic Electrodynamics (~75%)

Anchors: $SO_0(5,2)$ Lorentz subgroup of substrate symmetry - $SO(3,1)$ Lorentz $\subset$ $SO(5,2)$ substrate isometry (Wallach) - 4-current J^μ + 4-potential A^μ via substrate - Lorentz transformations on EM fields from substrate

Ch 8 Lagrangian + Hamiltonian EM (~70%)

Anchors: Yang-Mills action from T2477 - $L_{EM} = -(1/4) F_{\mu\nu} F^{\mu\nu}$ Yang-Mills Lagrangian - Action variation $\rightarrow$ Maxwell equations - Hamiltonian formulation + gauge fixing

Ch 9 Multipole Expansion + Scattering (~65%)

Anchor: Multipole moments + $\alpha^{\{\text{BST primary}\}}$ (T2476) - Multipole expansion: Y^l_m spherical harmonics - Scattering cross-sections at α -exponent BST primary - Klein-Nishina $\sigma \propto \alpha^2 = \alpha^{\{\text{rank}\}}$ (Vol 3 Ch 9)

Ch 10 EM in Matter (Dispersion, Optics) (~70%)

Anchor: Refractive-index BST primary ratios - Refractive indices at BST primary ratios - Dispersion relations from substrate Bergman propagation - Lorentz oscillator model + substrate K-type Casimir

Ch 11 Plasma Physics + MHD (~60%)

Anchor: Substrate plasma framework - Plasma frequency ω_p substrate-natural - MHD equations from substrate - Cross-link Vol 9 Ch 10 strongly correlated electrons (plasma analogy)

Ch 12 Bridge to Vol 3 + Vol 9 + Vol 8 (synthesis)

Anchor: Multi-volume integration - Vol 7 $\rightarrow$ Vol 3 (atomic EM phenomena) - Vol 7 $\rightarrow$ Vol 9 (condensed matter EM, photonics, plasma) - Vol 7 $\rightarrow$ Vol 8 (classical mechanics + electromagnetism unified)

BST primary appearances across Vol 7

All 5 BST primaries + N_{max} appear: - rank = 2: Klein-Nishina $\alpha^{\{\text{rank}\}}$ (Ch 9), Pin(2) substrate framework - $N_c = 3$: plasma 3D structure (Ch 11) - $n_c = 5$: Lamb shift $\alpha^{\{n_c\}}$ radiation (Ch 6) - $C_2 = 6$: substrate energy unit - $g = 7$: substrate field coupling - $N_{\text{max}} = 137$: $\alpha = 1/N_{\text{max}}$ universal (every chapter)

Methodology stack applied

- Cal #19 STANDING RULE: current ratified state per-chapter
- Cal #21 STANDING RULE: empirical + mechanism dual-gate
- Cal #99 META-theorem framing: substrate-derivation consequences
- Cal #22: PCAP-rate transcription discipline

Cross-volume dependencies (Vol 7)

Chapter	Vol 1	Vol 2	Vol 3	Vol 9
1 Substrate Reading	Ch 5+6+8	Ch 8	Ch 1	Ch 1
2 Maxwell	Ch 8 (T2477)	Ch 9 (T2478)	—	—
3 Electrostatics	Ch 5	Ch 8 (α)	Ch 4 (a_c)	—
4 Magnetostatics	Ch 6	—	—	Ch 6

Chapter	Vol 1	Vol 2	Vol 3	Vol 9
5 EM Waves	Ch 8	Ch 8	—	Ch 9
6 Radiation	—	Ch 8	Ch 8 ($\alpha^{\{n_C\}}$)	—
7 Relativistic	—	—	—	—
8 Lagrangian	Ch 8	—	—	—
9 Multipole	—	Ch 8 (T2476)	Ch 9	—
10 EM Matter	—	—	—	Ch 3, 9
11 Plasma	—	—	—	Ch 10
12 Bridge	All	All	All	All

Saturday Wave 3 advance cadence

Per sustained sub-PCAP: - INDEX [\[?\]](#) - Architectural Scaffold v0.1 (this) [\[?\]](#) - Vol 7 per-chapter v0.3 narratives (sustained Wave 3 push if time permits)

— Elie, Vol 7 Architectural Scaffold v0.1, 2026-05-23 Saturday